

North Bridge Academy — Network Assessment (Avram)

2026-06-23

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Date: Tue, Jun 23, 2026 (forwarded by Annie Crowder) **Author:** Avram, IT Coordinator

Internet Service

Detail	Value
Provider	Comcast coax (“cable modem”)
Download	~350 Mbps
Upload	~40 Mbps
Gateway	Up-to-date Comcast hardware
Issues	Brief midday connectivity losses at Comcast level. Need to isolate from WiFi radio issues.

WiFi / Router

- **System:** Ubiquiti Unifi
- **Access Points:** All reasonably current
- **Gateway:** Acceptable, though inexpensive upgrade to latest model may be justified
- **Management:** Unifi console under **WiLine** control (not the ISP — legacy from FastMetrics acquisitions). Avram has admin rights but not full system control.

- **WiLine status:** Running old management version on their hosted infrastructure. Barely charging anything (historical deal). Not particularly proactive — Avram found several APs needing manual software updates.
- **Equipment ownership:** North Bridge owns all equipment (as far as Avram knows)
- **Coverage:** Extensive in most classrooms. Power settings may need to be turned DOWN in central areas to avoid radio contention.

Issues Resolved by Avram

- Damaged cabling and small AP positioning that would have impacted Annie's Mac
- Mapped and tagged all network devices for visibility
- Monitored network remotely for several months
- One AP went offline for a few hours (unclear if work was happening in that office)

Current Assessment

No major WiFi radio connectivity problems at this time.

Definite Issues Requiring Resolution (Summer Window)

1. Router Firmware Update

- **Action:** Avram will trigger remotely in the next few days
- **Risk:** Should be routine and automatic

2. Unifi Network Management Upgrade

- **Blocker:** Only WiLine can perform this upgrade
- **Risk:** WiLine might not support it for their clients. If they won't, this is a problem — Avram's features are limited vs. even a home Unifi system.

3. UPS Backup — Partially Addressed

Component	UPS Status
Comcast Gateway	✓ On new APC UPS

Component	UPS Status
Router	✓ On new APC UPS
2nd Floor Switch	✗ NOT on battery backup — all 2nd floor APs go down immediately in outage
1st Floor Switch	✗ UPS has dead battery — replace battery (~cheap) or replace unit
Basement AP	✗ No battery backup

Larger Areas for Improvement

Better Internet: AT&T Fiber

- **Availability:** Deployed throughout Mill Valley
- **Benefit:** Huge improvement in speed and reliability vs. Comcast coax
- **Pricing:** Residential pricing is a bargain. Unknown if classified as commercial or educational.
- **Installation:** Bringing fiber from curb into building likely a major project (months, not weeks). May already be in building (doubtful).
- **Next step:** Start the inquiry. **Hesitancy noted** — needs discussion with Annie.

Redundant Failover: Starlink

- **Benefit:** Emergency failover internet
- **Complexity:** Starlink residential (easy, cheap) vs. commercial (different offering) — needs investigation
- **Compatibility:** Current Unifi setup supports 2 providers with automatic failover

Brewing Issue: IP Subnet Layout

- **Problem:** IP subnets were set up in a “cramped” way — actively running out of IPs as more devices join the LAN
- **Symptom:** “Can’t connect” for people entering the building at peak times

- **Temporary fix:** Avram extended with remaining room, but this is a band-aid
 - **Real fix:** Logical reconfiguration of entire layout. Phone subnet needs to be moved as well. This is a **project** — not trivial due to unfortunate initial decisions.
 - **Urgency:** OK at the moment, but a definite problem if network devices greatly expand. Must be done before class is in session.
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Avram's Summary

“Long story short nothing is urgent with network at the moment but I personally think some things should be addressed proactively over the summer, much safer than after.”

Available any time this week after noon to discuss network details by phone.

Annie's Take

Annie scheduled a 1-on-1 meeting with Frank (no Shannon, no Avram). Shannon is away this week. Annie wants to discuss “what their needs might be” — suggests she wants Frank's synthesis before bringing Avram back in.